

Generalisation Across Tumor Entities with a From-Scratch Residual-Encoder U-Net: A Baseline and an Analysis of Training–Validation Distribution Shift

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Abstract. Building a brain tumor segmentation model that transfers across patient populations remains unsolved in neuro-oncology. BraTS Generalizability Across Tumors (BraTS-GoAT) 2026 asks how well one model segments tumors from adult glioma, sub-Saharan African glioma, meningioma, brain metastasis and pediatric cohorts, without revealing the test set’s composition. We train a three-dimensional residual-encoder U-Net from scratch, with no external data or pretrained weights, and pair it with an analysis of what makes generalization hard here. Unsupervised fingerprinting of the training cases shows a pool dominated by a single tumor appearance group, while the evaluation data span five structurally different populations; we use this grouping to build stratified cross-validation folds. Per-fold, per-group evaluation shows segmentation quality markedly lower on structurally distinct minority groups than on majority glioma groups, with whole-tumor Dice falling from 0.95 to 0.89 and tumor-core Dice falling to 0.76 on the group lacking enhancing tumor. Two post-processing operations carried over from earlier BraTS editions, volume-based filtering of enhancing-tumor components and retention of only the largest whole-tumor component, work against the pooled evaluation metric used here; we report why we omit both. Our results point to the training-evaluation distribution mismatch, rather than network architecture, as the more likely limiting factor, compounded by an evaluation that weighs surface agreement as heavily as region overlap.

Keywords: Brain tumor segmentation · Generalization · Distribution shift · nnU-Net · Normalized surface distance · BraTS-GoAT.

1 Introduction

Segmentation of brain tumors from multi-parametric MRI (mpMRI) supports diagnosis, treatment planning and disease monitoring, and since 2012 BraTS has been the standard benchmark for the task [2,3,4]. BraTS-GoAT widens the problem: instead of one homogeneous population, a single algorithm must work across adult glioma, sub-Saharan African glioma [10], meningioma [6], brain metastasis [7] and pediatric [8,9] cohorts, including types unseen during training,

without knowing the private test set’s population mix. Ranking aggregates the Dice Similarity Coefficient (DSC) and Normalized Surface Distance (NSD) over three regions, enhancing tumor (ET), tumor core (TC) and whole tumor (WT), weighing boundary agreement roughly as heavily as overlap, and which labels appear also depends on tumor type: a case may lack necrotic core, edema, or enhancing tumor entirely. A model that assumes every scan contains every sub-region loses the most ground on populations furthest from the training majority.

We make three contributions: a from-scratch baseline within GoAT’s data rules, with its official validation-leaderboard performance; a quantitative analysis, per appearance group and per fold, of how the training pool’s composition diverges from GoAT’s stated evaluation populations, which we believe matters more for ranking than network choice; and an account of how that imbalance interacts with the surface-distance half of the metric and with post-processing habits inherited from earlier BraTS editions.

2 Methods

2.1 Challenge data and constraints

Each BraTS-GoAT 2026 training case ($n = 1351$) has four co-registered mpMRI sequences: native T1, contrast-enhanced T1 (T1Gd), T2, and T2-FLAIR, all skull-stripped and resampled to 1 mm^3 isotropic resolution. Reference masks carry three foreground labels: necrotic/non-enhancing core (NCR, 1), edema (ED, 2), enhancing tumor (ET, 3). Scoring uses three derived regions instead: ET, tumor core (TC = 1+3) and whole tumor (WT = 1+2+3).

Task rules forbid additional data or pretrained models; every component below trains from random initialisation on the provided set alone. Submitted containers are evaluated via the MedPerf framework [1]; the pooled corpus draws on adult glioma [2,3,4,5], meningioma [6], metastasis [7], pediatric [8,9] and sub-Saharan African [10] BraTS datasets.

2.2 Cohort fingerprinting and stratified folds

Case identifiers are opaque sequential integers, uninformative about population. To describe the data without external labels, we built a per-case fingerprint from the provided images and reference masks alone.

Each feature targets a specific, documented source of heterogeneity rather than an unconstrained search. Sub-region presence and log-volume encode the documented fact that some tumor types lack necrosis or enhancement entirely, which is the distributional claim under investigation. Enhancing-tumor volume fraction separates ET-dominant, low-infiltration tumors from diffuse gliomas. Brain bounding-box extent captures the pediatric-adult scale difference. Per-modality intensity percentiles capture scanner/acquisition variation, since GoAT pools data from five separately-sited sub-challenges.

Table 1. Unsupervised appearance groups ($n = 1351$) from k -means ($k = 5$, fixed a priori) over the features above. Per group: size, and for each ground-truth sub-region the *fraction of member cases in which it is present* (1.00 = always present, 0.00 = always absent). The final column is a qualitative summary for interpretability only; group identity is never used at inference. Groups 0 and 3 hold 90.7 % of training data; despite the shared label, Section 2.2 shows they separate by acquisition characteristics, not tumor presentation.

Group	n	NCR present	ED present	ET present	Consistent with
0	525	1.00	1.00	1.00	glioma (acq. subgroup A)
3	701	1.00	1.00	1.00	glioma (acq. subgroup B)
1	67	0.00	1.00	1.00	necrosis-free
2	33	0.82	1.00	0.00	non-enhancing
4	25	0.08	0.00	1.00	enhancing, edema-free

Features were standardized and clustered by k -means, seed fixed. $k = 5$ was set a priori to match the number of populations named in the challenge description, not chosen by optimising a clustering-quality criterion; the procedure is deterministic and reproducible.

The grouping is heavily imbalanced (Table 1): two groups hold roughly nine in ten cases. The other three are small and structurally distinct: no necrotic core, no enhancing tumor, and almost entirely enhancing tumor with little edema, respectively. The populations GoAT is designed to probe are therefore a small minority of the training pool.

Why groups 0 and 3 are two clusters, not one. Table 1 shows groups 0 and 3 share an identical presence pattern (NCR, ED, ET each $\approx 100\%$ in both), raising the question of why k -means splits them at all. Ranking features by standardized effect size (Cohen’s d) between the two groups shows the split is driven almost entirely by brain bounding-box extent, shaped by acquisition field-of-view, positioning and skull-stripping rather than pathology. The three bounding-box dimensions are the three largest effect sizes in the feature set ($d = 1.28, 0.76, 0.75$), well above any tumor-volume feature ($d \leq 0.47$) and, since every presence flag shows $d = 0.00$, far above the tumor-content features that track pathology directly; intensity percentiles are also elevated in group 0.

An ablation confirms this: removing bounding-box features collapses groups 0 and 3 into one cluster (524/525 and 701/701 merge), while groups 1, 2 and 4 are unaffected (67/67, 33/33, 25/25 unmixed). Bounding-box extent is therefore necessary and sufficient for this split.

The split shows up in performance too: group 0 outperforms group 3 on ET and TC Dice after controlling for tumor volume ($\Delta\text{DSC} = 0.030$, $p < 0.001$ ET; $\Delta\text{DSC} = 0.021$, $p = 0.002$ TC), though not on WT ($p = 0.31$, from a modest volume difference). Acquisition heterogeneity within a single nominal tumor population is therefore a measurable contributor to the generalization difficulty here, present even within the majority glioma portion and not only between glioma and minority populations.

Choice of cluster count. $k = 5$ was fixed, not chosen by an internal validity criterion. Silhouette analysis (one outlier excluded, else it forms a degenerate singleton at low k) is 0.545 at $k = 2$, peaks at $k = 3$ (0.622), then falls for $k \geq 4$ (0.499, 0.179, 0.185, 0.161, 0.167 for $k = 4-8$), favoring $k = 3$, which we reject: it merges groups 0 and 3 (an acquisition artifact, not a structural distinction) with 65/67 cases of group 1, whose separation is driven by near-total absence of necrotic core, one of the strongest ground-truth signals available. A coarser k conflates artifact with genuine structure. At the other end, $k = 6$ adds no additional structure: groups 1, 2 and 4 remain unchanged, and the extra cluster only fragments the 0/3 artifact further (664, 555, 6 cases), the last too small to be meaningful. $k = 5$ is the smallest count keeping every ground-truth-supported group distinct, and the largest before further splits stop finding real structure.

A random partition could push an entire minority group into one fold, so we stratified the five-fold split over these groups; Section 3.3 quantifies the benefit. Since folds are stratified on cluster label, each fold’s composition is near-identical by construction: deviation from the pooled mean is at most 0.10 percentage points for any cluster (largest 0.095 pp, non-enhancing group), confirming stratification held. We therefore report cluster composition once, pooled, rather than per fold. The clustering draws on nothing beyond the provided data and stays within task rules.

2.3 Preprocessing

Preprocessing followed the self-configuring nnU-Net pipeline [11]. Volumes were cropped to the non-zero brain region, intensities z -score normalized per modality and case, damping inter-scan variation while preserving anatomical contrast. Training patches were $128 \times 160 \times 112$ voxels, batch size 2, with a third of sampled patches centered on foreground.

2.4 Network architecture

The network is nnU-Net v2 ResEnc-M: a six-stage 3D U-Net, residual-block encoder, instance normalisation, LeakyReLU. 101.9M parameters, 88.6% in the encoder. Following the BraTS 2020 winning entry [12], it predicts the three evaluation regions directly ($WT \supseteq TC \supseteq ET$) rather than raw label classes, aligning the objective with the scoring definitions.

2.5 Loss function

Training minimises binary cross-entropy plus a Dice term computed over the pooled mini-batch, not per sample. About 2.4% of cases have no enhancing tumor, where per-sample Dice is degenerate since both prediction and target may be empty; pooling removes that instability and matches the challenge evaluation, which itself pools TP/FP/FN across the cohort before scoring each region.

Table 2. Post-processing ablation, pooled Dice and NSD over all 1351 held-out cross-validation predictions.

Variant	Region	DSC	NSD
baseline (hole-fill only)	ET	0.868	0.909
baseline (hole-fill only)	TC	0.908	0.886
baseline (hole-fill only)	WT	0.920	0.877
+ largest-WT-component	ET	0.857	0.895
+ largest-WT-component	TC	0.895	0.872
+ largest-WT-component	WT	0.912	0.867
+ ET volume filter	ET	0.870	0.908
+ ET volume filter	TC	0.906	0.884
+ ET volume filter	WT	0.920	0.876

2.6 Inference and post-processing

Inference uses sliding-window prediction with Gaussian patch weighting and mirroring test-time augmentation. Fold predictions are combined by averaging class probabilities before label assignment, not by voting on discrete labels.

Two post-processing operations common in earlier BraTS solutions were ablated on the fold-wise cross-validation predictions and both excluded, for different reasons.

Retaining only the largest WT component presupposes one confluent mass, true for most adult gliomas but not for metastatic or some pediatric cases, which are genuinely multifocal; keeping only the largest component then discards every other lesion. The ablation (Table 2) confirms this is harmful across all three regions: ET DSC fell 0.012, TC 0.012, WT 0.008, with NSD moving the same way.

ET volume-filtering produced a different result: negligible aggregate effect ($\Delta\text{DSC} = +0.001$) across all 1351 cases, which might suggest it is harmless. Broken down by appearance group, the aggregate masks two opposing effects that grow with threshold. Group 1 (necrosis-free) loses ET DSC, from -0.013 at 25 voxels to -0.117 at 500, consistent with real small lesions being removed. Group 2 (non-enhancing) instead gains, $+0.061$ to $+0.273$, matching a filter that suppresses false-positive ET on cases that structurally have none. Majority groups (0, 3) show a small, growing cost ($-0.004/-0.013$ at the highest threshold). We omit this filter not for uniform harm, but because no single threshold is neutral or beneficial across every group at once.

The one operation we keep is morphological hole filling on WT: it can only add voxels within an existing component and can never remove a separate lesion, making it safe under the pooled metric.

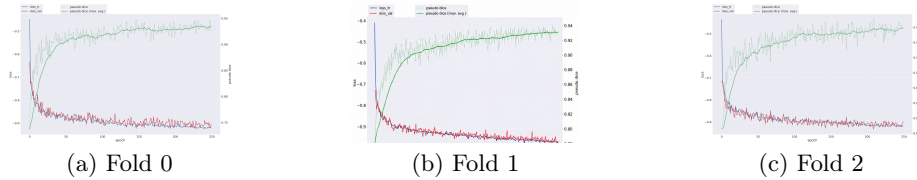
2.7 Implementation details

All models trained from random initialisation on a single NVIDIA T4 GPU. Table 3 lists the full configuration; every setting is the nnU-Net ResEnc-M default except batch-Dice and the reduced epoch budget. No hyperparameter search was performed.

Training curves for folds 0, 1 and 2 (Fig. 1) show loss and pseudo-Dice flattening well before the epoch budget was exhausted. EMA validation Dice improves

Table 3. Complete training configuration. All values are nnU-Net v2 ResEnc-M defaults unless marked as modified.

Setting	Value
Optimiser	SGD, momentum 0.99, Nesterov polynomial decay,
Learning-rate schedule	0.01 $\rightarrow$ 0
Batch size	2
Patch size	$128 \times 160 \times 112$
Normalisation	z-score, per modality, per case
Target spacing	$1.0 \times 1.0 \times 1.0$ mm
Loss	Dice + binary cross-entropy
Batch-Dice	enabled (modified from default)
Epochs	250 (modified from default 1000)
Framework	nnU-Net v2 (2.5.1), PyTorch 2.10.0

**Fig. 1.** Training loss and pseudo-Dice for three representative folds, shown for reference. All three plateau well before the 250-epoch budget is exhausted.

by under 0.003 over the final fifty epochs, so the reduced schedule ended after convergence, not during active optimisation. The same early plateau appears in the one other monitored fold regardless of stopping point. Folds 3 and 4 did not reach the full 250 epochs; the per-cluster figures in Section 3.2 use whichever checkpoint was available for each. Folds 0, 1 and 2 completed the full schedule and use final-epoch checkpoints directly.

Preprocessing, architecture, optimisation and inference were identical across folds, with fixed cross-validation splits throughout, so all results reproduce from a fixed seed.

3 Results

3.1 Leaderboard performance

Table 4 reports performance on the official BraTS-GoAT 2026 validation leaderboard for three ensemble configurations.

Both ensembles beat the single fold on TC and WT but not ET. The three-fold ensemble, built only from folds trained to completion, matches or exceeds the five-fold ensemble on every NSD column despite fewer models, while five-fold holds a marginal TC-Dice edge. We submit three-fold, since NSD and DSC weigh equally in the ranking and it is the more consistent of the two, plausibly because fold 4’s under-converged checkpoint dilutes rather than diversifies the average.

Under Dice both configurations rank regions the same way: WT best, ET worst. NSD reverses this, which we attribute to boundary geometry rather than prediction quality: the WT surface is long and runs through diffuse edema, harder to meet a fixed tolerance along than the compact ET rim.

Table 4. Performance on the official validation leaderboard. DSC is the Dice Similarity Coefficient and NSD the Normalized Surface Distance, each reported for enhancing tumor (ET), tumor core (TC) and whole tumor (WT); higher is better for both. The five-fold configuration includes fold 4, stopped before the full training schedule completed (Section 2.7). Best value per column in bold.

Configuration	DSC			NSD		
	ET	TC	WT	ET	TC	WT
Single fold	0.7765	0.7874	0.8425	0.5276	0.4677	0.4319
3-fold ensemble	0.7745	0.7942	0.8540	0.5290	0.4721	0.4443
5-fold ensemble	0.7744	0.7954	0.8532	0.5270	0.4719	0.4426

Table 5. Fold-wise cross-validation DSC and NSD (1 mm tolerance), mean $\pm$ std., pooled $n = 1351$ across five folds ($n \approx 270$ per fold).

Fold	DSC (mean $\pm$ std.)			NSD @ 1mm (mean $\pm$ std.)		
	ET	TC	WT	ET	TC	WT
0	0.875 $\pm$ 0.191	0.904 $\pm$ 0.183	0.919 $\pm$ 0.129	0.913 $\pm$ 0.194	0.883 $\pm$ 0.213	0.874 $\pm$ 0.160
1	0.869 $\pm$ 0.186	0.896 $\pm$ 0.189	0.922 $\pm$ 0.130	0.911 $\pm$ 0.184	0.876 $\pm$ 0.218	0.882 $\pm$ 0.156
2	0.880 $\pm$ 0.173	0.922 $\pm$ 0.131	0.928 $\pm$ 0.119	0.914 $\pm$ 0.177	0.902 $\pm$ 0.170	0.886 $\pm$ 0.151
3	0.859 $\pm$ 0.206	0.905 $\pm$ 0.163	0.914 $\pm$ 0.124	0.899 $\pm$ 0.206	0.880 $\pm$ 0.195	0.869 $\pm$ 0.164
4	0.858 $\pm$ 0.209	0.910 $\pm$ 0.160	0.919 $\pm$ 0.118	0.908 $\pm$ 0.203	0.888 $\pm$ 0.195	0.875 $\pm$ 0.155
Pooled	0.868 $\pm$ 0.194	0.908 $\pm$ 0.166	0.920 $\pm$ 0.124	0.909 $\pm$ 0.193	0.886 $\pm$ 0.199	0.877 $\pm$ 0.157

Table 5 reports fold-wise cross-validation DSC and NSD (1 mm tolerance), mean $\pm$ std. These held-out predictions (pooled $n = 1351$) score consistently higher than the leaderboard above, since cross-validation draws from the training distribution ($\sim 91\%$ glioma) while Synapse validation is deliberately heterogeneous. Per-fold variation is modest for every region (e.g. WT DSC 0.914–0.928), matching the near-identical fold composition from Section 2.2.

3.2 Per-group and per-fold performance

Table 6 reports Dice on held-out cases by appearance group (Table 1), the direct test of whether performance tracks the training-pool imbalance.

WT Dice is markedly lower on the three minority groups (0.919, 0.896, 0.894) than on either majority glioma group (0.954, 0.943), on the exact population GoAT probes. TC Dice on the non-enhancing and enhancing/edema-free groups falls to about 0.76, roughly seventeen to nineteen points below either glioma group. Pooled ET Dice for the non-enhancing group is 0.000: ground-truth ET is absent for every case in this group by construction, so any false-positive ET prediction anywhere in the pooled set drives the denominator positive against a zero numerator.

Table 7 breaks WT down per fold, across all five folds now that held-out predictions are available for every fold. Within the two majority groups, per-fold values are stable (within roughly 0.02 WT Dice). The enhancing, edema-free group (cluster 4, $n = 25$) is the exception: WT Dice on fold 0 is 0.29, well below its other folds (0.55–0.74). Case-level inspection shows this is driven by cases with no predicted foreground on a subset of this group’s smallest lesions (under roughly 600 ground-truth voxels); 8 of the 25 cluster-4 cases have at least one zero-prediction fold. We attribute this to the reduced step_size used for this

Table 6. Pooled Dice on held-out validation cases, by appearance group, across all five folds ($n = 1351$). Pooled Dice sums TP/FP/FN across cases before dividing, matching the challenge metric. Predictions were re-scored offline at `step_size=0.5`, not the submitted configuration’s 0.25 (Section 2.7). Groups follow the ordering of Table 1.

Group	n	WT	TC	ET
0 glioma A	525	0.954	0.954	0.926
3 glioma B	701	0.943	0.934	0.918
1 necrosis-free	67	0.896	0.907	0.924
2 non-enhancing	33	0.919	0.764	0.000
4 enhancing, edema-free	25	0.894	0.766	0.891

Table 7. Per-fold whole-tumor Dice by appearance group, all five folds ($n = 1351$ held-out cases total). Values are per-case Dice averaged within each fold-group cell, not pooled; see Table 6 for pooled figures at the same `step_size=0.5` setting.

Group	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
0 glioma A	0.9506	0.9527	0.9506	0.9392	0.9411
3 glioma B	0.9261	0.9257	0.9296	0.9244	0.9239
1 necrosis-free	0.8629	0.7790	0.8735	0.8235	0.7828
2 non-enhancing	0.8399	0.9298	0.9272	0.7866	0.9050
4 enhancing, edema-free	0.2857	0.5548	0.5330	0.5202	0.7430

offline re-scoring pass rather than to the model itself: sparser sliding-window coverage is the setting most directly linked to missing small, spatially compact lesions, and the submitted configuration’s denser `step_size=0.25` was chosen in part for this reason (Section 2.7).

3.3 Effect of stratification

To quantify what appearance-group stratification contributes, we compared our stratified split against 2000 random unstratified five-fold splits over the same group labels (Table 8).

For the rarest group, random assignment yields an expected minimum of 2.6 cases/fold against a guaranteed 5 under stratification, and 1.9% of random splits leave a fold entirely empty of that group, making per-group evaluation impossible there. Stratification is thus a prerequisite for Table 6’s analysis, not an incidental choice.

4 Discussion

4.1 Findings supported by our experiments

Table 6 supports the central claim directly: WT Dice is consistently lower on structurally distinct minority groups than on either majority glioma group, and non-enhancing-group TC Dice trails either glioma group by roughly seventeen to nineteen points. Since all groups share training pool, architecture and optimisation, the difference reflects training representation, not model capacity. This

Table 8. Stratified versus random five-fold assignment, over 2000 random splits. “Min/fold” is the smallest number of cases of that group appearing in any validation fold. Stratification guarantees a positive minimum by construction; random assignment does not.

Group		n	Stratified min/fold	Random mean min/fold	Random $P(\text{empty fold})$
0	glioma A	525	105	95.8	0.0%
3	glioma B	701	140	130.7	0.0%
1	necrosis-free	67	13	9.4	0.0%
2	non-enhancing	33	6	3.8	0.1%
4	enhancing, edema-free	25	5	2.6	1.9%

per-group evaluation is only possible because folds were stratified (Table 8); random assignment would leave a fold empty of the rarest group in a non-negligible fraction of splits.

Table 4 shows averaging folds improves TC and WT but not ET, and that including under-converged folds does not help: folds differ only in training subset, so averaging cancels boundary disagreement but not a systematically missed component. The training curves also show convergence before the reduced epoch budget was exhausted, so the reported overlap is not primarily an artifact of truncated training.

4.2 Interpretation

We find no evidence that architecture is the limiting factor here, though we test only one; the pattern in Table 6 is more readily explained by the structural mismatch between a training pool dominated by one appearance and an evaluation set spanning five populations, worsened by a metric weighing surface distance as heavily as overlap. This is not only a between-population issue: it appears within the majority glioma portion itself (Section 2.2), where two acquisition-differentiated subgroups show a Dice gap surviving volume control.

Two consequences follow. ET varies most across tumor types and is where the baseline is weakest, so effort directed there likely yields more improvement than further effort on the already-strong WT region; and because ET is missing in a non-trivial share of cases, methods that assume its presence are penalized accordingly, which is why we encode the opposite assumption in both pooled-batch Dice and the population-dependent ET-filter treatment (Section 2.6). Surface distance is half the ranking signal and consistently the lower score; inference-time choices sharpening boundary agreement without changing overlap offer gains at no extra training cost.

4.3 Limitations

The minority groups are small: the three structurally distinct groups hold between 25 and 67 cases (Table 1), so per-group estimates carry wide uncertainty and per-fold variation is relatively large. The groups are unsupervised constructs, correlated with but not identical to documented source cohorts; we do not claim exact correspondence.

Pooling introduces an additional caveat: per-case failures on minority groups are invisible in a pooled figure. Section 3.2 documents this directly for the smallest group: 8 of 25 enhancing, edema-free cases have at least one fold in which no foreground was predicted despite nonzero ground truth, concentrated on the smallest lesions. A pooled-only view can therefore understate how unreliable predictions are on individual minority-group cases.

The post-processing decisions of Section 2.6 are backed by ablation on our own predictions rather than metric reasoning alone, but three further choices rest on a proxy quantity rather than a full retraining comparison. Table 8 quantifies the sampling risk a random split would carry, not a trained-model comparison requiring retraining every fold under both regimes. Pooled-batch Dice (Section 2.5) and hole filling (Section 2.6) are similarly adopted on metric-alignment grounds, supported indirectly by the overall results rather than a dedicated ablation.

Code Availability Code for data conversion, cohort fingerprinting, training, inference and the analysis scripts reported in this paper is available at <https://github.com/Goober-ly/NeuroAI-BRATS-GoAT2026>.

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